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Patent Public Search | Text View

United States Patent Application Publication

20250282687

Kind Code

A1

Publication Date

September 11, 2025

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Visual Communications

Abstract

A persistent virtual area that supports establishment of respective presences of communicants operating respective network nodes connected to the virtual area even after all network nodes have disconnected from the virtual area is maintained. A presence in the virtual area is established for a user of a Public Switched Telephone Network (PSTN) terminal device. Transmission of data associated with the virtual area to the PSTN terminal device.

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Appl. No.: 19/214343

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 18336795 20230616 parent-grant-document US 12312274 child US 19214343

parent US continuation 17242605 20210428 parent-grant-document US 11785056 child US 18336795

parent US continuation 16120026 20180831 parent-grant-document US 11025679 child US 17242605

parent US continuation 15168481 20160531 parent-grant-document US 10069873 child US 16120026

parent US continuation 13165729 20110621 parent-grant-document US 9357025 child US 15168481

Publication Classification

Int. Cl.: **C04B35/119** (20060101); **C04B35/03** (20060101); **C04B37/02** (20060101); **H01L23/15** (20060101); **H04L65/1069** (20220101); **H04L65/1104** (20220101); **H04L65/65** (20220101); **H04L67/131** (20220101); **H04L67/52** (20220101); **H04L67/54** (20220101); **H04W4/021** (20180101)

U.S. Cl.:

CPC **C04B35/119** (20130101); **C04B35/03** (20130101); **C04B37/021** (20130101); **C04B37/025** (20130101); **H01L23/15** (20130101); **H04L65/1069** (20130101); **H04L65/1104** (20220501); **H04L65/65** (20220501); **H04L67/131** (20220501); **H04L67/52** (20220501); **H04L67/54** (20220501); **H04W4/021** (20130101); C04B2235/3222 (20130101); C04B2235/5436 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of prior U.S. patent application Ser. No. 12/694,126, filed Jan. 26, 2010, the entirety of which is incorporated herein by reference. [0002] This application also relates to the following patent applications, the entirety of each of which is incorporated herein by reference: U.S. Provisional Patent Application No. 61/449,059, filed Mar. 3, 2011; U.S. Provisional Patent Application No. 61/373,914, filed Aug. 16, 2010; U.S. patent application Ser. No. 12/825,512, filed Jun. 29, 2010; U.S. patent application Ser. No. 12/509,658, filed Jul. 27, 2009; U.S. patent application Ser. No. 12/418,270, filed Apr. 3, 2009; U.S. patent application Ser. No. 12/354,709, filed Jan. 15, 2009; U.S. application Ser. No. 12/630,973, filed on Dec. 4, 2009; U.S. patent application Ser. No. 11/923,629, filed Oct. 24, 2007; and U.S. patent application Ser. No. 11/923,634, filed Oct. 24, 2007.

BACKGROUND

[0003] When face-to-face communications are not practical, people often rely on one or more technological solutions to meet their communications needs. These solutions typically are designed to simulate one or more aspects of face-to-face communications. Traditional telephony systems enable voice communications between callers. Instant messaging (also referred to as “chat”) communications systems enable users to communicate text messages in real time through instant message computer clients. Some instant messaging systems additionally allow users to be represented in a virtual environment by user-controllable graphical objects (referred to as “avatars”). Interactive virtual reality communication systems enable users in remote locations to communicate and interact with each other by manipulating their respective avatars in virtual spaces.

Description

DESCRIPTION OF DRAWINGS

[0004] FIG. 1 is a block diagram of an example of a network communications environment that includes a first client network node, a second client network node, a PSTN device, and a virtual

environment creator.

[0005] FIGS. 2-4 are diagrammatic views of graphical user interface examples.

[0006] FIG. 5 is a flow diagram of an example of a method of interfacing a PSTN device terminal user to a virtual area.

[0007] FIG. 6A is a block diagram of an example of a network communications environment that supports virtual area based telephony communications.

[0008] FIG. 6B is a flow diagram of an example virtual area based telephony method.

[0009] FIG. 7 is a block diagram of an example of the network communications environment of FIG. 6A.

[0010] FIG. 8A is a diagrammatic view of a virtual area and an associated graphical interface for initiating a virtual area based telephony communication.

[0011] FIG. 8B is a diagrammatic view of a virtual area and an associated graphical interface for managing a virtual area based telephony communication.

[0012] FIG. 8C is a diagrammatic view of a virtual area and an associated graphical interface for managing a virtual area based telephony communication.

[0013] FIGS. 9A-9C are diagrammatic views of graphical interfaces for getting a PSTN terminal device user into a virtual area.

[0014] FIG. 10A is a block diagram of an example of a network communications environment that supports virtual area based telephony communications.

[0015] FIG. 10B is a flow diagram of an example virtual area based telephony method.

[0016] FIG. 11A is a block diagram of an example of a network communications environment that supports virtual area based telephony communications.

[0017] FIG. 11B is a flow diagram of an example virtual area based telephony method.

DETAILED DESCRIPTION

[0018] In the following description, like reference numbers are used to identify like elements.

Furthermore, the drawings are intended to illustrate major features of exemplary embodiments in a diagrammatic manner. The drawings are not intended to depict every feature of actual embodiments nor relative dimensions of the depicted elements, and are not drawn to scale.

I. DEFINITION OF TERMS

[0019] A “communicant” is a person who communicates or otherwise interacts with other persons over one or more network connections, where the communication or interaction may or may not occur in the context of a virtual area. A “user” is a communicant who is operating a particular network node that defines a particular perspective for descriptive purposes.

[0020] A “computer” is any machine, device, or apparatus that processes data according to computer-readable instructions that are stored on a computer-readable medium either temporarily or permanently. A “computer operating system” is a software component of a computer system that manages and coordinates the performance of tasks and the sharing of computing and hardware resources. A “software application” (also referred to as software, an application, computer software, a computer application, a program, and a computer program) is a set of instructions that a computer can interpret and execute to perform one or more specific tasks. A “data file” is a block of information that durably stores data for use by a software application.

[0021] The term “computer-readable medium” refers to any tangible, non-transitory medium capable storing information (e.g., instructions and data) that is readable by a machine (e.g., a computer). Storage devices suitable for tangibly embodying such information include, but are not limited to, all forms of physical, non-transitory computer-readable memory, including, for example, semiconductor memory devices, such as random access memory (RAM), EPROM, EEPROM, and Flash memory devices, magnetic disks such as internal hard disks and removable hard disks, magneto-optical disks, DVD-ROM/RAM, and CD-ROM/RAM.

[0022] A “window” is a visual area of a display that typically includes a user interface. A window typically displays the output of a software process and typically enables a user to input commands

or data for the software process. A window that has a parent is called a “child window.” A window that has no parent, or whose parent is the desktop window, is called a “top-level window.” A “desktop” is a system-defined window that paints the background of a graphical user interface (GUI) and serves as the base for all windows displayed by all software processes.

[0023] A “data sink” (referred to herein simply as a “sink”) is any of a device (e.g., a computer), part of a device, or software that receives data.

[0024] A “data source” (referred to herein simply as a “source”) is any of a device (e.g., a computer), part of a device, or software that originates data.

[0025] A “network node” (also referred to simply as a “node”) is a junction or connection point in a communications network. Examples of network nodes include, but are not limited to, a terminal, a computer, and a network switch. A “server” network node is a host computer on a network that responds to requests for information or service. A “client network node” is a computer on a network that requests information or service from a server.

[0026] A Uniform Resource Identifier (URI) is a string of characters that identifies a network resource.

[0027] A “network resource” is anything that can be identified by a uniform resource identifier (URI) and accessed over a network, including an electronic document, an image, a source of information, a service, operators and operands of a mathematical equation, classes, properties, numeric values, and a collection of other resources.

[0028] A “network connection” is a link between two communicating network nodes. A “connection handle” is a pointer or identifier (e.g., a uniform resource identifier (URI)) that can be used to establish a network connection with a network resource. A “network communication” can include any type of information (e.g., text, voice, audio, video, electronic mail message, data file, motion data stream, and data packet) that is transmitted or otherwise conveyed from one network node to another network node over a network connection.

[0029] Synchronous conferencing refers to communications in which communicants participate at the same time. Synchronous conferencing encompasses all types of networked collaboration technologies, including instant messaging (e.g., text chat), audio conferencing, video conferencing, application sharing, and file sharing technologies.

[0030] A “communicant interaction” is any type of direct or indirect action or influence between a communicant and another network entity, which may include for example another communicant, a virtual area, or a network service. Examples of types of communicant communications include communicants communicating with each other in realtime, a communicant entering a virtual area, and a communicant requesting access to a resource from a network service.

[0031] “Presence” refers to the ability and willingness of a networked entity (e.g., a communicant, service, or device) to communicate, where such willingness affects the ability to detect and obtain information about the state of the entity on a network and the ability to connect to the entity.

[0032] A “realtime data stream” is data that is structured and processed in a continuous flow and is designed to be received with no delay or only imperceptible delay. Realtime data streams include digital representations of voice, video, user movements, facial expressions and other physical phenomena, as well as data within the computing environment that may benefit from rapid transmission, rapid execution, or both rapid transmission and rapid execution, including for example, avatar movement instructions, text chat, realtime data feeds (e.g., sensor data, machine control instructions, transaction streams and stock quote information feeds), screen shares, and file transfers.

[0033] A “virtual area” (also referred to as an “area” or a “place”) is a representation of a computer-managed space or scene. Virtual areas typically are one-dimensional, two-dimensional, or three-dimensional representations; although in some examples a virtual area may correspond to a single point. Oftentimes, a virtual area is designed to simulate a physical, real-world space. For example, using a traditional computer monitor, a virtual area may be visualized as a two-dimensional graphic

of a three-dimensional computer-generated space. However, virtual areas do not require an associated visualization. A virtual area typically refers to an instance of a virtual area schema, where the schema defines the structure and contents of a virtual area in terms of variables and the instance defines the structure and contents of a virtual area in terms of values that have been resolved from a particular context.

[0034] A “persistent virtual area” is a virtual area that persists even after all communicants have disconnected from the virtual area. The state of a persistent virtual area is preserved so that it can be restored the next time a communicant connects to the virtual area.

[0035] A “virtual area application” (also referred to as a “virtual area specification”) is a description of a virtual area that is used in creating a virtual environment. A virtual area application typically includes definitions of geometry, physics, and realtime switching rules that are associated with one or more zones of the virtual area.

[0036] A “virtual area based communications application” is a client communications application that integrates realtime audio communications (and potentially other realtime communications, e.g., video, chat, and realtime other data stream) with visual presentations of interactions in a virtual area.

[0037] A “virtual environment” is a representation of a computer-managed space that includes at least one virtual area and supports realtime communications between communicants.

[0038] A “zone” is a region of a virtual area that is associated with at least one switching rule or governance rule. A “switching rule” is an instruction that specifies a connection or disconnection of one or more realtime data sources and one or more realtime data sinks subject to one or more conditions precedent. A switching rule controls switching (e.g., routing, connecting, and disconnecting) of realtime data streams between network nodes communicating in the context of a virtual area. A governance rule controls a communicant's access to a resource (e.g., an area, a region of an area, or the contents of that area or region), the scope of that access, and follow-on consequences of that access (e.g., a requirement that audit records relating to that access must be recorded). A “renderable zone” is a zone that is associated with a respective visualization.

[0039] A “position” in a virtual area refers to a location of a point or an area or a volume in the virtual area. A point typically is represented by a single set of one-dimensional, two-dimensional, or three-dimensional coordinates (e.g., x, y, z) that define a spot in the virtual area. An area typically is represented by the three-dimensional coordinates of three or more coplanar vertices that define a boundary of a closed two-dimensional shape in the virtual area. A volume typically is represented by the three-dimensional coordinates of four or more non-coplanar vertices that define a closed boundary of a three-dimensional shape in the virtual area.

[0040] A “spatial state” is an attribute that describes where a user has presence in a virtual area. The spatial state attribute typically has a respective value (e.g., a zone_1D value) for each of the zones in which the user has presence.

[0041] A “communication state” is an attribute that describes a state of a respective communication channel over which a respective one of the communicants is configured to communicate.

[0042] As used herein, the term “includes” means includes but not limited to, the term “including” means including but not limited to. The term “based on” means based at least in part on.

II. VIRTUAL AREA BASED TELEPHONY COMMUNICATIONS

[0043] The examples that are described herein provide virtual area based telephony systems and methods that enable a user of a Public Switched Telephone Network (PSTN) terminal device to participate in a virtual area based communications. Some examples enable the PSTN terminal device user to connect to audio data that is associated with a persistent virtual area. In this way, the PSTN terminal device user can participate in realtime communication sessions with other communicants in the virtual area, retrieve data that is stored in association with the virtual area, or store data in association with the virtual area. Some examples enable a communicant in a persistent virtual area to bring (or “get”) a PSTN terminal device user into a virtual area based

communication session simply by calling the PSTN terminal device. Some examples enable a PSTN terminal device user to establish a presence in or interaction with data associated a virtual area or a particular zone of a virtual by calling a telephone number associated with the virtual area or zone.

[0044] FIG. 1 shows an embodiment of an exemplary network communications environment **10** that includes a first client network node **12**, a second client network node **14**, a PSTN device **16**, and a virtual environment creator **18** that are interconnected by a network (not shown) that support the transmission of a wide variety of different media types (e.g., text, voice, audio, video, and other data) between network nodes.

[0045] Each of the first client network node **12** and the second client network node **14** includes input/output (I/O) hardware, a processor, and a computer-readable memory that stores a respective instance **20, 22** of at least one virtual-area-enabled communications application that is executable by the processor. Each of the network nodes **12, 14** has a respective set of one or more sources and an exemplary set of one or more sinks. Exemplary sources include an audio source (e.g., an audio capture device, such as a microphone), a video source (e.g., a video capture device, such as a video camera), a chat source (e.g., a text capture device, such as a keyboard), a motion data source (e.g., a pointing device, such as a computer mouse), and other sources (e.g., file sharing source or a source of a customized real-time data stream). Exemplary sinks include an audio sink (e.g., an audio rendering device, such as a speaker or headphones), a video sink (e.g., a video rendering device, such as a display monitor), a chat sink (e.g., a text rendering device, such as a display monitor), a motion data sink (e.g., a movement rendering device, such as a display monitor), and other sinks (e.g., a printer for printing shared files, a device for rendering real-time data streams different from those already described, or software that processes real-time streams for analysis or customized display).

[0046] The communications applications **20, 22** respectively operating on the client nodes **12, 14** typically include software and hardware resources which, together with administrative policies, user preferences (including preferences regarding the exportation of the user's presence and the connection of the user to the area applications **34** and other communicants), and other settings, define a local configuration that influences the administration of realtime connections with other network nodes. Among the software components typically executing on the client network nodes **12, 14** are a user interface component and a browser component. The browser component provides a set of web browsing functions, including browser functions, document viewing functions, and data downloading functions. The user interface component generates a graphical user interface that interfaces the user to the realtime communications and network browsing functionalities of the browser component. The browser component may be integrated into the communications applications **20, 22** or it may be implemented by a separate browser component (e.g., a plug-in) that exposes an API through which the communications applications **20, 22** may call methods that are available from the browser component, including browsing methods, document viewing methods, and data downloading methods.

[0047] The PSTN terminal device **16** is any device that communicates over the PSTN, including mobile devices (e.g., mobile phones and portable computing devices, such as tablet and notebook computers) and fixed-line telephones.

[0048] In the illustrated embodiment, the virtual environment creator **18** includes at least one server network node(s) **28** that provides a network infrastructure service environment **30** that manages sessions of the first and second client nodes **12, 14** and the PSTN terminal device **16** in one or more virtual areas **32** in accordance with respective virtual area applications **34**. Each of the virtual area applications **34** is hosted by a respective one of the virtual areas **32** and includes a description of the respective virtual area **32**. Communicants respectively operating from the client network nodes **12, 14** connect to the virtual area applications **34** through the communications applications **20, 22**. The server network node(s) **28** also includes components (e.g., a Voice eXtensible Markup

Language (VXML) interpreter, a speech recognition engine, a text to speech synthesizer, and a Dual Tone Multi-Frequency (DTMF) decoder) that enable a user of the PSTN terminal device **16** to connect to the virtual area applications **34** through one or more PSTN modalities.

[0049] The network infrastructure service environment **30** typically includes one or more network infrastructure services that cooperate with the communications applications **20, 22** in the process of establishing and administering network connections between the client nodes **12, 14** and other network nodes, including the PSTN terminal device **16**. Among the network infrastructure services that are included in an exemplary embodiment of the network infrastructure service environment **30** are an account service, a security service, an area service, a rendezvous service, and an interaction service. The structure, operation, and components of such an embodiment of the network infrastructure service environment **30** are described in U.S. patent application Ser. No. 12/825,512, filed Jun. 29, 2010.

[0050] The communications applications **20, 22**, the area applications **34**, and the network infrastructure service environment **30** together provide a platform (referred to herein as “the platform”) that administers the realtime connections with network nodes in a communication context that is defined by an instance of a virtual area subject to a set of constraints that control access to the virtual area instance.

[0051] In the illustrated embodiment, the communications applications **20, 22** operating on the first and second client network nodes **12, 14** present respective spatial visualizations **36, 38** (or views) of the virtual areas **32** in accordance with data received from the network infrastructure service environment **30**. The communications applications **20, 22** also provide respective interfaces for receiving commands from the communicants and providing a spatial interface that enhances the realtime communications between the communicants. The spatial visualizations **36, 38** include respective graphical representations **52, 54** (referred to herein as “avatars” or “sprites”) of the client communicants in spatial relation to a graphical representation **58** of a virtual area. The spatial visualizations **36, 38** also may include other objects. Examples of such props include a view screen object **55** for application sharing, a table object **57** for file sharing, and a conferencing object **59** for initiating telephone calls to PSTN terminal devices. The user of the PSTN terminal device **16** also is represented in the virtual areas by a respective avatar **56**. In some examples, the avatars **62, 64, 70** are moved about the virtual areas **32** based on commands that are input by the communicants at their respective network nodes **12, 14, 16**.

[0052] The spatial visualizations **36, 38** on the client nodes **12, 14** are presented in respective windows **42, 44** that are generated by the virtual area enabled communications applications **20, 22**. The communication application windows **42, 44** typically are displayed on respective “desktops” or other system-defined base windows **48, 50** on the display hardware of the client nodes **12, 14**.

[0053] The platform tracks communicants' realtime availabilities and activities across the different communication contexts that are defined by the area applications **34**. This information is presented to the communicants in the form of realtime visualizations that enable the communicants to make more informed network interaction decisions (e.g., when to interact with a contact) and encourages the communicants to initiate interactions with other communicants and to join contexts (e.g., an ongoing conversation between communicants) of which the communicants otherwise would not have been aware. In some embodiments, the realtime visualization includes visual cues as to the presence and activities of the communicants in the contexts of the server applications. The presentation of these visual cues typically depends on one or more of governance rules associated with the virtual areas **32**, administrative policies, and user preferences (including preferences regarding the exportation of the user's presence and the connection of the user to areas and other communicants), which may define tiered relationship based predicates that control access to presence information and/or resources on a zone-by-zone basis.

[0054] In some embodiments, the server network node(s) **28** remotely manages client communication sessions and remotely configures audio and graphic rendering engines on the client

network nodes **12, 14**, as well as switching of data streams by sending instructions (also referred to as definitions) from the remotely hosted area applications **34** to the client network nodes in accordance with the stream transport protocol described in U.S. patent application Ser. No. 12/825,512, filed Jun. 29, 2010. In some of these embodiments, the server network node(s) **28** send to each of the client nodes **12, 14** provisioning messages that configure the client nodes **12, 14** to interconnect respective data streams between active ones of their complementary sources and sinks in accordance with switching rules specified in the area applications **34**. In some embodiments, server network node(s) **28** remotely configures the communications applications **20, 22** to establish realtime data stream connections between the first and second client network nodes **12, 14** and other network nodes sharing the virtual area **32** based on the positions of the communicants' avatars in the virtual area **32**.

[0055] The server network node(s) **28** also manages connection of the PSTN terminal device **16** to the virtual area **32** so that a PSTN terminal device user can participate in virtual area based communications (e.g., communicate with one or more other communicants who are in the virtual area **32**, or retrieve data from or store data in the virtual area **32**).

[0056] FIG. **2** shows an exemplary graphical user interface **70** that is generated by an example of the communications application **20** in a window **61** on a display of the client network node from which a user of the client application (“Art” in this example) is operating. The graphical user interface **70** includes a people panel **65**, a viewer panel **66**, a people interaction toolbar **67**, an audio interaction toolbar **68**, and a set of panel view controls **69**. The panel view controls **69** include a people panel button **71** for opening and closing the people panel **65**, a Chat panel button **73** for opening and closing a Chat panel, and a viewer panel button **75** for opening and closing the viewer panel **66**.

[0057] The people panel **66** depicts the realtime availabilities and activities of some or all of Art's contacts across different communication contexts. In the example shown in FIG. **2**, the people panel **66** shows Art's communicants segmented into two virtual area groups **72, 74** and a contacts group **76**. The virtual area groups **72, 74** correspond to each of the area applications **34** of which Art is a member with respect to which at least one of Art and Art's contacts is present. The contacts group **76** contains all or a selected portion of Art's contacts that are not represented in any of the virtual area groups. The first virtual area group **72** of communicants is contained within a section **78** (labeled with a header bar entitled “Zone 1”) that identifies all the communicants who have a presence in the area application “Application 1”. The second virtual area group **74** of communicants is contained within a section **80** (labeled with a header bar entitled “Zone 2”) that identifies all the communicants who have a presence in the area application “Application 2”. The contacts group **76** of communicants is contained within a section **82** (labeled with a header bar entitled “Contacts”) that identifies all of Art's contacts who are not shown in any of the first and second virtual area groups **72, 74** (i.e., they either are not members of or not present in any of Zone **1** and Zone **2**).

[0058] In the example shown in FIG. **2**, the virtual area sections **78, 80** contain the graphical representations (avatars) of the communicants (including at least one of Art or Art's contacts) who currently have presence in the respective area applications **34**, and the contacts section **82** contains graphical representations (avatars) of all of the remaining ones of Art's contacts that are not present in or not members of any of Zone **1** and Zone **2**. In the illustrated example: Art and Beth are members of server Zone **1**; Art, Carl, Dan, and Ed are members of server Zone **2**; and Fran and Garth are not members of server Zone **1** nor server Zone **2**. In this example, Fran and Garth are contacts of Art who are not presented in Zone **1** or Zone **2**.

[0059] Each communicant is represented graphically by a respective circular sprite that is labeled with a respective user name of the communicant (i.e., “Art,” “Beth,” “Carl,” “Dan,” “Ed,” “Fran,” and “Garth”). Each sprite also may be associated with a respective status line that includes additional information about the communicant. In some embodiments, each status line can include

one or more of the following information: location of presence (e.g., a server application or a zone of that sever application); availability (e.g., busy, idle); a status message (e.g., “Out of the office next Wednesday”); and the name of the client node from which the communicant is operating (e.g., “workstation 1” or “mobile phone”). In some embodiments, the ordering of the spatial positions (e.g., from top to bottom) of the communicant avatars in each of the sections **78, 80, 82** is alphabetical by user name. In other embodiments, the spatial positions of the communicant avatars in each of the server application sections **78, 80** are ordered in accordance with the temporal ordering of the communicants in terms of the times when the communicants established their respective presences with the server applications. The spatial positions of the communicant avatars in the contacts section **82** may be sorted alphabetically by user name, according to frequency of contact, according to recentness of contact, or according to other sorting or filtering criteria.

[0060] The activities of the communicants in the contexts of the area applications **34** may be inferred from the states of various communication channels over which the respective communicants are configured to communicate. The states of the communication channels are shown in the graphical user interface **70** by visual cues that are depicted in association with the graphical representations of the communicants in the sections **78, 80, 82**. For example, the “on” or “off” state of a communicant's local speaker channel is depicted by the presence or absence of a headphones graphic **90** on the communicant's sprite. When the speakers of the communicant who is represented by the sprite are on, the headphones graphic **90** is present (see sprites Art, Carl, and Dan) and, when the communicant's speakers are off, the headphones graphic **90** is absent. The “on” or “off” state of the communicant's microphone is depicted by the presence or absence of a microphone graphic **92** on the communicant's sprite. When the microphone is on, the microphone graphic **92** is present; and, when the microphone is off, the microphone graphic **92** is absent. A PSTN terminal device user is represented in the spatial visualization by a sprite that includes a telephone handset graphic **91** (see sprite Ed). The headphones graphic **90**, the microphone graphic **92**, and the headset graphic **91** provide visual cues of the activity states of the communicant's sound playback and microphone devices. In addition, the current activity state of a communicant's microphone channel is indicated by a dynamic visualization that lightens and darkens the communicant's avatar in realtime to reflect the presence or absence of audio data on the microphone channel. Thus, whether or not their local speakers are turned on, communicants can determine when another communicant is speaking by the “blinking” of the coloration of that communicant's avatar. The activity state of a communicant's text chat channel is depicted by the presence or absence of the hand graphic **94** adjacent the communicant's sprite (see sprite Beth). Thus, when a communicant is transmitting text chat data to another network node the hand graphic **94** is present, and when a communicant is not transmitting text chat data the hand graphic **94** is not present. In some embodiments, text chat data is transmitted only when keyboard keys are depressed, in which case the visualization of the communicant's text channel appears as a flashing on and off of the hand graphic **94**.

[0061] In the example shown in FIG. 2, members of an area application are able to receive the visual cues of the communicant activities occurring in the context of that area application whether or not the member is present. Thus, the graphical user interface **70** that is presented to Art shows visual cues indicating the communication channel states of the communicants present in Zone **1** (where Art is present) and the communication channel states of the communicants present in Zone **2** (where Art is not present).

[0062] The audio interaction toolbar **68** includes a headphone control **84** that enables Art to toggle on and off the local speakers of the client network node, and a microphone control **86** that enables Art to toggle on and off the local microphone of the client network node.

[0063] Additional details regarding embodiments of the people panel **65** are described in U.S. Provisional Patent Application No. 61/373,914, filed Aug. 16, 2010, and U.S. patent application Ser. No. 12/354,709, filed Jan. 15, 2009.

[0064] The viewer panel **66** includes a navigation area **110** and a display area **112**.

[0065] The navigation area **110** includes forward and back buttons **114**, a location bar **116**, a Go button **118**, and a reload button **120**. The forward and back buttons **114** enable a user to traverse a navigation stack of uniform resource identifier (URI) addresses (e.g., a linked list of previously visited URLs). The location bar **116** allows a user to specify a URI address of a network resource, and the Go button **118** invokes one or more browser functions on the client network node to navigate to the specified URI address and render the network resource at the specified URI address in the display area **112**. The reload button **120** invokes one or more browser functions on the client network node to reload the graphic representation of the network resource currently displayed in the display area **112**.

[0066] The display area **112** contains the rendered depictions of network resources located at the URI address specified in the navigation area **110**. In the example shown in FIG. **14**, the viewer panel **66** is in the browser view mode and shows a rendered view of the network resource (a web page in this example) that is located at the URL <https://www.sococo.com/home.php>, as indicated in the location bar **116**. In the illustrated example, the display area **110** shows a web page that includes a header section **122**, a top navigation bar **124**, a side navigation bar **126**, a contents section **128**, a notices section **130**, and a navigation links section **132**.

[0067] The people interaction toolbar **67** includes a Chat button **98** and an Invite button **102**.

Selection of the Chat button **98** opens a Chat panel **140** (see FIG. **4**) that enables Art to initiate a chat with other communicants who are present in the area application where Art is present (i.e., Zone **1** in the illustrated example). Selection of the Invite button **102** opens an Invite window that enables Art to invite one or more communicants to a selected virtual area location (e.g., an area application or zone within that area application). Additional details regarding embodiments of the methods and functions invoked by the Chat button **98** and the Invite button **102** are described in U.S. patent application Ser. No. 12/354,709, filed Jan. 15, 2009, and U.S. Provisional Patent Application No. 61/373,914, filed Aug. 16, 2010.

[0068] In addition to the control and panel elements of the graphical user interface **70** (e.g., the people panel **65**, the viewer panel **66**, the people interaction toolbar **67**, the audio interaction toolbar **68**, and the panel view controls **71**, **73**, **75**), the graphical user interface **70** includes a Share button **375** and a set **373** of Viewer Panel control buttons, including a Map button **376**, a Browse button **378**, and four View Screen buttons **380-386**. The Share button **375** initiates a screen share of the contents of the display area **112** of the viewer panel **66** in connection with a view screen in a virtual area. These contents include renderings of any information that is received by the browser component in connection with the network resource identified in the location bar **116**, and a document or application that is being shared by the user in connection with a view screen object in a virtual area. The Map button **376** sets the view presented in the viewer panel **66** to a map view of the virtual area. The Browse button **378** sets the view presented in the viewer panel **66** to a browser view. Each of the four View Screen buttons **380-386** sets the viewer panel **66** to display the content the content being shared in connection with a corresponding one of the view screens in the virtual area.

[0069] FIG. **3** shows an example of the graphical user interface **70** in the Map view mode presenting in the viewer panel **66** a rendered view of an example of a virtual area that includes a first zone (Zone **1**) and a second zone (Zone **2**). The first zone is located at the location SococoHQ/Zone**1**, as indicated in the location bar **110**. This address indicates that Zone **1** is a zone within the area SococoHQ.

[0070] Each of the communicants who is present in the virtual area is represented graphically by a respective avatar that corresponds to the communicant's avatar that is shown in the people panel **65**. The virtual area is represented graphically by a two-dimensional top view of a rectangular space. In some examples, the communicants' sprites automatically are positioned in predetermined locations (or "seats") in the virtual area when the communicants initially enter the virtual area.

[0071] The virtual area includes four view screen props **388, 390, 392, 394** and a table prop **396**. Communicants interact with the props by selecting them with an input device (e.g., by single-clicking on the props with a computer mouse, touch pad, touch screen, or the like). The view screen props **388-394** are associated with application sharing functionality of the platform that enables communicants to share applications operating on their respective client network nodes. The application sharing functionality is invoked by activating a view screen (e.g., by single-clicking the view screen object with an input device). In some embodiments, the platform provides visual cues that indicate whether or not a communicant is sharing an application over an application sharing channel. For example, in response to a communicant's selection of the view screen prop, the communicant's sprite automatically is moved to a position in the graphical representation of the virtual area that is adjacent the view screen prop. The position of a communicant's sprite adjacent the view screen prop indicates that the communicant currently is sharing or is about to share an application with the other communicants in the virtual area. In addition, the avatar of each communicant who is viewing a shared application is depicted with a pair of "eyes" to indicate that the represented communicants are viewing the content being shared in connection with the view screen props. The graphical depiction of view screen prop is changed depending on whether or not an active application sharing session is occurring. For example, the depicted color of the view screen may change from a brighter color during an active application sharing session to a darker color when there is no application sharing taking place. Examples of the application sharing process are described in connection with FIGS. 26-28 of U.S. patent application Ser. No. 12/354,709, filed Jan. 15, 2009, and in U.S. patent application Ser. No. 12/418,270, filed Apr. 3, 2009.

[0072] The table prop **396** is associated with file share functionality of the platform that enables communicants to upload computer data files to server storage in association with the virtual area and to download data files that are associated with the virtual area from the server storage to the respective client network nodes. In example shown in FIG. 3, there are two document objects **398, 400** that are associated with the table prop **396**. The document objects **398, 400** are linked to respective documents that have been shared in the virtual area and stored in server storage. Any of the document objects **398, 400** may be selected by a communicant (e.g., by double-clicking the document object **190** with an input device, such as a computer mouse) to initiate downloading of the associated document to the communicant's client network node. Additional details regarding the structure, function, and operation of the table prop **396** may be obtained from U.S. patent application Ser. No. 12/354,709, filed Jan. 15, 2009.

[0073] In the Map view mode, the navigational controls of the graphical user interface **70** allow the user to traverse a path through the virtual environment in accordance with a navigational model that is tied to the underlying spatial hierarchy of virtual area locations and objects (e.g., props) within the locations. The network infrastructure service environment records the path traversed by the user. In some embodiments, the network infrastructure service environment records a history that includes a temporally ordered list of views of the virtual area locations that are presented to the user as the user navigates through the virtual area. Each view typically corresponds to a view of a respective renderable zone of the virtual area. In these embodiments, the navigation controls enable the user to move to selected ones of the zones in the history. The navigation controls also include a graphical representation of a depth path that shows the location in the spatial hierarchy that corresponds to the user's current view of the virtual area. In some embodiments, the graphical representation of the depth path includes a respective user-selectable link to a respective view of each of the preceding levels in the spatial hierarchical model of the virtual area above the current view. The back button **369** corresponds to a backward control that enables the user to incrementally move backward to preceding ones of the zones in the history of the zones that were traversed by the user. The forward button **371** corresponds to a forward control that enables the user to incrementally move forward to successive ones of the zones in the history of the zones that were

traversed by the user. Some examples additionally include a placemarks button that activates a placemarking control for storing links to zones and a placemark navigation control for viewing a list of links to previously placemarked zones. In response to user selection of the placemarking control, a placemark is created by storing an image of the location shown in the current view in association with a hyperlink to the corresponding location in the virtual area. In response to a user selection of the placemark navigation control, a placemarks window is presented to the user. The placemarks window includes live visualizations of all locations that have been placemarked by the user. Each of the images in the placemarks window is associated with a respective user-selectable hyperlink. In response to user selection of one of the hyperlinks in the placemarks window, a view of the virtual area corresponding to the location associated with the selected hyperlink is automatically displayed in the browsing area of the graphical user interface **70**. Some examples include home button corresponds to a control that returns the user to a view of a designated “home” location in the virtual environment. Additional details regarding the structure, function, and operation of examples of the navigation controls are described in U.S. patent application Ser. No. 12/354,709, filed Jan. 15, 2009.

[0074] FIG. **4** shows an example of the graphical user interface **70** when the people panel **65**, a Chat panel **402**, and the viewer panel **66** are open.

[0075] Activating the Chat panel button **73** or the Chat button **98** opens the Chat panel **140**. When the Chat panel button **73** is activated, the Chat panel **140** opens to show a chat interface for a persistent virtual chat area for interactions occurring in connection with a respective virtual area. In the example shown in FIG. **7**, Art activated the Chat panel button **73** at the time he was present in Zone **1**; therefore, the Chat panel **140** shown in FIG. **4** contains the persistent virtual chat area for text chat interactions occurring in connection with Zone **1**. When the Chat button **98** is activated, on the other hand, the Chat panel **140** opens to show a chat interface for a persistent personal virtual area for interactions between Art and a selected one of the communicants. Examples of personal virtual areas are described in U.S. patent application Ser. No. 12/509,658, filed Jul. 27, 2009.

[0076] The chat interface of the Chat panel **140** includes a chat log area **142**, a text box **144**, and a Send button **146**. The Chat panel **402** also includes a minimap view of a virtual area (Zone **1**). In this example, the user may enter text messages in the text box **144** and activate the Send button **146** to transmit the text messages to the other communicants who are present in the virtual area.

[0077] The user may enter text messages in the text box **144** and transmit the text messages to the other communicants who are in the same chat session by selecting the Send button **146**. The chat log area **142** displays a log of current and optionally prior events that are associated with Application **1**. An exemplary set of events that may be displayed in the chat log area **142** include: text messages that the user has exchanged with other communicants in Application **1**; changes in the presence status of communicants in Application **1**; changes in the speaker and microphone settings of the communicants in the Application **1**; and the status of the props (discussed below), including references to any applications and data files that are shared in connection with the props. In the illustrated embodiments, the events are labeled by the communicant's name followed by content associated with the event (e.g., a text message) or a description of the event.

[0078] The Chat panel **140** provides a context for organizing the presentation of the events that are displayed in the chat log area **142**. For example, in the illustrated embodiment, each of the displayed events is labeled with a respective tag that visually correlates with the appearance of the sprite of the communicant that sourced the displayed event. In particular, each of the events that is sourced by a particular one of the communicants is labeled with a respective icon **148**, **150**, **152**, **154** with a visual appearance (e.g., color-code, or design pattern) that matches the visual appearance of that communicant's sprite. In this example, the color of the icons **148**, **152** matches the color of the body of Art's sprite, and the color of the icon **150**, **154** matches the color of the body of Beth's sprite.

[0079] FIG. 5 shows an example of a method by which an example of the virtual environment creator **18** interfaces a PSTN terminal device user to a virtual area.

[0080] In accordance with the method of FIG. 5, the virtual environment creator **18** maintains a persistent virtual area that supports establishment of respective presences of communicants operating respective network nodes connected to the virtual area even after all network nodes have disconnected from the virtual area (FIG. 2, block **160**). The virtual area typically has one or more zones. One or more of the virtual area and the one or more zones of the virtual area typically is associated with at least one respective telephone number (e.g., a direct inward dialing (DID) number or a caller ID number).

[0081] The virtual environment creator **18** establishes a presence in the virtual area **32** for a user of the PSTN terminal device **16** (FIG. 2, block **162**). The presence of the PSTN terminal device user may be established in the virtual area or in a particular zone of the virtual area. The presence of the PSTN terminal device user may be established, for example, in response to a request by a communicant who is present in the virtual area to call the PSTN terminal device **16** or in response to receipt of a telephone call from the PSTN terminal device **16**.

[0082] The virtual environment creator **18** administers transmission of data associated with the virtual area to the PSTN terminal device **16** (FIG. 2, block **164**). The transmitted data typically is any type of data (e.g., audio data or text data) that can be rendered by the PSTN terminal device **16** or can be converted by conversion device or service (e.g., a telephony service provider) into a format (e.g., audio data via a text-to-speech synthesizer) that can be rendered by the PSTN terminal device **16**.

[0083] In some examples, the virtual environment creator **18** establishes a respective presence in the virtual area **32** for a communicant operating a respective one of the client network nodes **12**, **14**. The virtual environment creator **18** manages establishment of a realtime audio connection between the client network node **12** and the PSTN terminal device **16** in association with the virtual area **32**. The virtual environment creator **18** also typically stores persistent records describing interactions between the PSTN terminal device user and the communicant operating the client network node **12** in the virtual area **32**.

[0084] In some examples, the virtual environment creator **18** initiates the realtime audio connection with the PSTN terminal device **16** in response to a request received from the client network node **12**. The request typically includes a telephone number that is associated with the PSTN terminal device **16**. In the process of initiating the realtime audio connection, the virtual environment creator **18** typically transmits the telephone number to a PSTN service provider (e.g., an Internet Telephony Service Provider (ITSP)), which administers connections with the PSTN terminal device **16**. The virtual environment creator **18** also typically transmits a telephone number (e.g., a caller ID number) that associated with the virtual area to the PSTN service provider. In some examples, the virtual area is associated with a spatial visualization, and the request is received from the client network node **12** in connection with the spatial visualization. In some of these examples, the spatial visualization includes a graphic representation of a room and a graphic representation of audio call connection object (e.g., a conferencing object) in the room, and the request is received in connection with the audio call connection object.

[0085] In some examples, the virtual environment creator **18** initiates the realtime audio connection in response to a request received in connection with the PSTN terminal device **16**. In some of these examples, the virtual area **32** or a particular zone of the virtual area **32** is associated with a telephone number (e.g., a Direct Inward Dialing number (DID)), and the request includes the telephone number. The virtual environment creator **18** establishes a presence for the PSTN terminal device user in the area or zone that is associated with the telephone number.

[0086] In some examples, the virtual environment creator **18** manages a current realtime communication session between the communicant and the PSTN terminal device user in the virtual area **32**. In this process, the virtual environment creator **18** typically specifies locations of graphical

representations of the user and the communicant in spatial relation to a graphical representation of the virtual area **32**. In some of these examples, based on entry of another communicant operating a second client network node **14** into the virtual area **32**, the virtual environment creator **18** administers establishment of a respective audio connection between the other client network node **14** and each of the first client network node **12** and the PSTN terminal device **16**.

[0087] In some examples, the virtual environment creator **18** derives audio data that is transmitted to the PSTN terminal device **16** from one or more realtime audio data streams that are respectively received from one or more client network nodes being operated by one or more respective communicants who are present in the virtual area **32**. In some of these examples, the virtual environment creator **18** receives each of the one or more realtime audio data streams from a respective one of the one or more client network nodes over a respective peer-to-peer network connection. In this process, the virtual environment creator **18** may receive respective realtime audio data streams from multiple client network nodes, in which case the virtual environment creator **18** derives the audio data from the realtime audio data streams by mixing the realtime data streams. The virtual environment creator **18** typically transmits the derived audio data to the PSTN terminal synchronously with receipt of the one or more realtime audio data streams in a realtime communication session between the user of the PSTN terminal device and the one or more respective communicants.

[0088] In some examples, the virtual environment creator **18** stores data in association with the virtual area **32** such that the stored data remains associated with the virtual area **32** independent of any communicant presence in the virtual area. For example, the virtual environment creator **18** may store a recording of one or more realtime audio streams (or a mix of such audio streams) received from communicants who are present in virtual area. Subsequently, the virtual environment creator **18** may transmit audio data derived from the stored recording to one or more of the PSTN terminal device **16** and the client network nodes **12**, **14** whenever they connect to the virtual area and select an associated audio playback object in the virtual area.

[0089] FIG. **6A** shows an example **170** of the network communications environment **10** that supports virtual area based telephony communications in which a communicant in a virtual area is able to bring a PSTN terminal device user into the virtual area by calling the user's PSTN terminal device from the virtual area.

[0090] The network communications environment **170** includes first and second area servers **172**, **174** and a Voice Over Internet Protocol (VOiP) gateway server **176**. Each of the first and second area servers **172**, **174** executes a respective instance of the area service that administers a respective virtual area in which communicants can interact with each other and with data stored in association with the virtual areas. In the illustrated example, the first area server **172** creates a first virtual area in which a client network node user (represented by a sprite **178**) and a first PSTN terminal device user (represented by a sprite **180**) are present. The second area server **174** creates a second virtual area in which second and third PSTN terminal device users (represented by sprites **182**, **184**) are present. In this example, the first, second, and third PSTN device users are referred to as VOiP gateway clients because they interface with the first and second virtual areas through respective gateway client applications that are executed by the VOiP gateway server **176**. The VOiP gateway server **76** communicates with the user's PSTN terminal devices **186**, **188**, **190** (e.g., a fixed-line telephone, a mobile telephone, and a SIP phone) through an Internet Telephony Service Provider (ITSP) **192**. The ITSP **192** communicates with the PSTN terminal devices **186-190** over the PSTN. In the illustrated example, the area servers **172**, **174** communicate with an optional billing server **194** that manages user billing accounts to which telephony and other charges are applied.

[0091] FIG. **6B** shows an example virtual area based telephony method in which a client network node user (e.g., the user of client network node **12**) initiates a virtual area based telephony communication with a user of a PSTN terminal device (e.g., PSTN device **186**).

[0092] In accordance with the method of FIG. **6B**, the user of the client network node **12** initiates a

realtime audio connection with the PSTN terminal device **186** (FIG. 6B, block **200**). In this process, the client network node user logs into the first virtual area through the first area service instance running on the first area server **172**. The first area server **172** creates an instance of the first virtual area and establishes a presence for the user in the first virtual area. After a presence for the user has been established, the user may initiate a telephone call to the user of the PSTN terminal device **186** through a conferencing object in the first virtual area.

[0093] In response to the user input received in connection with the conferencing object, the first area service instance validates the permissions and account balance of the user of the client network node **12** to determine whether or not the user is allowed to establish the virtual area based telephony session with the PSTN terminal device (FIG. 6B, block **202**). The first area service instance will not establish the virtual area based telephony session if the governance rules associated with the virtual area do not allow the client network node user to call the PSTN terminal device user; or if the user's account balance is insufficient to make the call, in which case the user may be prompted to authorize an increase to the account balance.

[0094] After the permissions and account balance of the user of the client network node **12** have been validated, the first area service instance sends a VoipInitiate request to the VOiP gateway service running on the VOiP gateway server **176** (FIG. 6B, block **204**).

[0095] In response to receipt of the VoipInitiate request, the VOiP gateway server **176** spawns an instance of a gateway client for the PSTN terminal device **186** (FIG. 6B, Block **206**). In some examples, the gateway client corresponds to a modified version of the virtual area based client application **20** running on the client network node **12**. In this way, the gateway client can be configured for virtual area based communications by the area service instance running on the area server **172**.

[0096] The spawned VOiP gateway service sends to the ITSP **192** a Session Initiation Protocol (SIP) Invite message requesting a session with the PSTN terminal device **186** (FIG. 6B, block **208**). The ITSP **192** completes the SIP Invite by sending a SIP Ack to the VOiP gateway service.

[0097] After the SIP Invite has been completed, a RealTime Protocol (RTP) session is established between the VOiP gateway service and the ITSP **192** (FIG. 6B, block **210**). The ITSP **192** also establishes a PSTN connection with the PSTN terminal device **186**. Audio data is transmitted between the ITSP **192**, the VOiP gateway service, and the gateway client instance over the RTP session (FIG. 6B, block **212**). The audio data is transmitted between the gateway client instance and the client network node **12** (FIG. 6B, block **214**).

[0098] In some examples, the VOiP gateway server **176** includes a translation engine (e.g., a DTMF character decoder, a VXML interpreter, or a speech recognition engine) that translates signals received from the ITSP **192** into commands for interacting in the virtual area. In some examples, the translation engine translates DTMF characters into commands for navigating the PSTN terminal device user to different zones of the virtual area, interacting with objects in the virtual area, or controlling one or more of the PSTN terminal device user's speaker and microphone channels.

[0099] In some examples, the VOiP gateway server **176** includes a translation engine for converting DTMF characters or VXML data received from the ITSP **192** into chat text and for converting chat text into speech or VMXL. A Short Message Service (SMS) bridge or gateway server exchanges the chat text related data with the ITSP **192**. In this way, the user of the PSTN terminal device **186** is able to participate in a text chat conversation with another communicant in the virtual area or access text chat messages that are stored in association with the virtual area.

[0100] In some examples, the gateway client instance and the client network node **12** establish a peer-to-peer communication session in accordance with the STRAW stream transport protocol described in U.S. patent application Ser. No. 12/825,512, filed Jun. 29, 2010, which defines channels for transmitting media records and Sococo Definition Architecture (SODA) records each containing one or more SODA definitions (e.g., maintenance definitions, client provisioning

definitions, definitions of 3D rendering assets, and definitions of Realtime Differential Streams (RDS), such as avatar motion checkpoints). In this way, the area service can configure the gateway clients and the communications applications executing on the client network nodes for virtual area based communications in essentially the same way.

[0101] FIG. 7 shows an example **400** of the network communications environment **170** (see FIG. 6A) that includes an example **402** of the VOiP gateway server **170**. The VOiP gateway server **402** includes a VOiP gateway server process **404** for communicating with the area server **172**, a respective VOiP gateway client process **406** for each of the client network nodes in a virtual area, and an interprocess pipe **408** through which the gateway server process **404** communicates with the gateway client processes **406**.

[0102] The gateway server process **404** includes a VOiP gateway service **410** that includes a MicroSork kernel **412**, which corresponds to a stripped version of the SORK kernel described in U.S. application Ser. No. 12/630,973, filed on Dec. 4, 2009, that includes the message handling functions but omits the audio mixing functions. The gateway server process **404** also includes a telephony service **414** that includes a user agent **416** for communicating with the ITSP **192**, a conversation manager **418** for communicating with the VOiP gateway client process **406** over the interprocess pipe **408**, and SIP and RTP libraries **420**, **422** (e.g., reSIProcate libraries) for building SIP and RTP communication stacks.

[0103] Each VOiP gateway client process **406** includes a VOiP gateway client service **424**, a SORK kernel **426**, and an audio service manager **428**, an outgoing interprocess pipe data handler **430** (e.g., a reSIProcate producer), and an ingoing interprocess pipe data handler **432** (e.g., a reSIProcate consumer). Examples of the SORK kernel **426** and the audio service manager **428** are described in U.S. application Ser. No. 12/630,973, filed on Dec. 4, 2009.

[0104] FIGS. 8A-8C show an exemplary spatial visualization of a zone **220** (Zone 1) of a virtual area and respective graphical user interfaces for managing calls to PSTN terminal devices. The virtual area zone **220** contains four viewscreen objects **222**, **224**, **226**, **228**, a table object **230**, and a conferencing object **232**. The virtual area zone **220** also contains four sprites **234**, **236**, **238**, and **240** representing respective communicants (Dan, Beth, Art, and Carl) who are present in the virtual area zone **220**. As explained below, any of the communicants in the virtual area zone **220** can bring (or “get”) a PSTN terminal device user into a communication session in the virtual area zone **220** simply by calling the user's PSTN terminal device via the conferencing object **232**.

[0105] Referring to FIG. 8A, in response to activation of the conferencing object **232** by a respective one of the communicants, the area service instance administering the virtual area zone **220** transmits a specification of a dialer interface **242** to the communicant's client network node, which renders a graphic representation of the dialer interface **242** on the communicant's display. The dialer interface **242** includes a Phone Number input box **244** for entering the telephone number of the target PSTN terminal device, a Name input box **246** for entering or displaying the name of the target user of the target PSTN terminal device, a keypad **248** for entering the telephone number in the Phone Number input box **244**, a Call button **250** for submitting the telephone number displayed in the Phone Number input box **244** to the area service, and a Cancel button **252** for closing the dialer interface **242**.

[0106] Referring to FIG. 8B, upon receipt of the telephone number of the target PSTN terminal device from the client network node, the area service administers the establishment of a virtual area based telephony session with the target PSTN terminal device. While the virtual area based telephone session is being established, the area service instance administering the virtual area zone **220** transmits a specification of a call-in-progress interface **254** to the communicant's client network node, which renders a graphic representation of the call-in-progress interface on the communicant's display. The call-in-progress interface **254** includes an End Call button **256** that allows the user to terminate the call establishment process. The call-in-progress interface **254** also includes a graphic representation **258** of the callee that indicates that a call is being established. In

the illustrated example, the graphic representation consists of dashed lines that depict the callee in multiple positions, where the dashed lines suggest that the callee is not yet present in the virtual area zone and the multiple depicted positions corresponds to a spatial representation of a “ringing” sound that typically is used in telephony calls to indicate that the call has gone through but has not yet been answered. A similar graphical representation **260** of the callee is shown in the virtual area zone **220**.

[0107] Referring to FIG. **8C**, after the call with the target PSTN terminal device has been established, the area service instance administering the virtual area zone **220** transmits a specification of a call-connected interface **262** to the communicant's client network node, which renders a graphic representation of the call-connected interface **262** on the communicant's display. The call-connected interface **262** includes an End Call button **264** that allows the user to terminate the call, a Keypad button for displaying the dialer interface **242**, and a close button **268** for closing the call-connected interface **262**. The call-connected interface **262** also includes a graphic representation **270** of the callee that indicates that a call is connected. In the illustrated example, the graphic representation **270** of the callee shows a complete depiction of a sprite that corresponds to the graphic representations of the other communicants in the room, except that the callee's sprite includes a graphic representation of a telephone indicating that the callee is connected to the virtual area zone **220** via a PSTN terminal device. A similar graphical representation **272** of the callee is shown in the virtual area zone **220**.

[0108] In the example described above, any of the communicants in the virtual area zone **220** is able to bring a PSTN terminal device user into the virtual area zone **220** by entering a telephone for the callee and clicking on the Call button **250** in the dialer interface **242**.

[0109] FIGS. **9A-9C** show alternative graphical interfaces for bringing a PSTN terminal device user into a virtual area or a particular zone of a virtual area. Like the dialer interface **242**, each of these graphical interfaces includes a respective Get button **275**, **277**, **279** that enables a user to get a target callee into a virtual area with one click. FIG. **9A** shows a People interface **274** that displays an alphabetically ordered list of all of the user's contacts and members of the current virtual area or zone in which the user is present who are callable (i.e., has a known telephone number). FIG. **9B** shows a Recent Calls interface **276** that displays a list of all of the callees that the user has called in order from most recently called to least recently called. FIG. **9C** shows a Favorites interface **278** that displays a list of callees that the user has designated as a Favorite ordered by call frequency from most frequently called to least frequently called. Each of the interfaces **274-278** enables the user to scroll through the displayed list of callees, select a target callee, and click the Get button to bring the target callee into the virtual area from which the user is calling (e.g., West Conference). In some examples, the user is able to select multiple target callees from the displayed list and bring all of the target callees into the current virtual area or zone with a single click of the Get button.

[0110] FIG. **10A** shows an example **280** of the network communications environment **10** that supports virtual area based telephony communications in which a user of a PSTN terminal device can establish a presence in or interact with data associated with a virtual area or a particular zone of a virtual area by calling a dial-in telephone number that is associated with the virtual area or zone.

[0111] The network communications environment **280** includes first and second area servers **172**, **174**, a VOiP gateway server **176**, a SIP proxy server **286**, a SIP registrar server **288**, and an optional asterisk server **290**. Each of the first and second area servers **172**, **174** executes a respective instance of the area service that administers a respective virtual area in which communicants can interact with each other and with data stored in association with the virtual areas. In the illustrated example, the first area server **172** creates a first virtual area in which client network node users (represented by sprites **178**, **282**) are present. The second area server **174** creates a second virtual area in which a third client network node user (represented by sprite **284**) is present. The VOiP gateway server **76** communicates with a user's PSTN terminal device **186** through an Internet Telephony Service Provider (ITSP) **192**, which communicates with the PSTN terminal devices over

the PSTN. The SIP proxy server **286** communicates with the VOiP gateway server **176**, the ITSP **192**, the SIP registrar server **288**, and the asterisk server **290**. The asterisk server **290** provides additional services to the PSTN terminal device users, including voice mail services, auto attendant services, and announcement services.

[0112] FIG. **10B** shows an example virtual area based telephony method in which a user of a PSTN terminal device (e.g., PSTN device **186**) initiates a virtual area based telephony communication with a virtual area or a particular zone of a virtual area by dialing the dial-in telephone number that is associated with the virtual area or zone.

[0113] In accordance with the method of FIG. **10B**, each of the area services running on the area servers **174**, **176** select a respective VOiP gateway server to host a respective dial-in telephone number that is associated with a respective virtual area or zone (FIG. **10B**, block **292**). In the illustrated example, the area service running on the area server **172** selects the VOiP gateway server **176** to host a first dial-in number that is associated with the virtual area or zone in which the client network node users **178**, **282** are present; and the area service running on the area server **174** also selects the VOiP gateway server **176** to host a second dial-in telephone number that is associated with the virtual area or zone in which the client network node user **282** is present.

[0114] The VOiP gateway server **176** registers each of the dial-in telephone numbers with the SIP proxy registrar server **288** (FIG. **10B**, block **294**).

[0115] In response to receipt of a call from the PSTN terminal device **186** to a respective one of the dial-in telephone numbers, the ITSP **192** sends a SIP Invite message to the SIP proxy server **286** (FIG. **10B**, block **296**). The SIP proxy server **286** requests the network address (e.g., a URI, such as a URL) of the VOiP gateway server that is associated with the dial-in telephone number from the SIP registrar server **288** (FIG. **10B**, block **298**). The SIP registrar server **288** returns the network address of the VOiP gateway server to the SIP proxy server **286** (FIG. **10B**, block **300**).

[0116] The SIP proxy server **286** sends the SIP Invite message to the VOiP gateway server **176**. In response, the VOiP gateway server **176** spawns an instance of a gateway client for the PSTN terminal device **186** and sends a notification of the in-bound telephone call to the area service hosting the virtual area or zone associated with the dial-in telephone number (FIG. **10B**, block **303**). The area services sends the session keys (also referred to as “credentials”) for the virtual area or zone to the VOiP gateway server **176** (FIG. **10B**, block **304**).

[0117] The VOiP gateway service completes the SIP Invite by sending a SIP Ack to the ITSP **192** (FIG. **10B**, block **304**). The VOiP gateway service and the ITSP **192** establish a RTP session (FIG. **10B**, block **308**). Audio data is transmitted between the ITSP **192**, the VOiP gateway service, and the gateway client instance over the RTP session (FIG. **10B**, block **310**). The audio data is transmitted between the gateway client instance and the client network nodes (FIG. **10B**, block **312**).

[0118] In some examples, the virtual environment creator **18** establishes a presence for a PSTN terminal device user in a virtual area or a particular zone of the virtual area in response to receipt of a telephone number that is associated with the virtual area or a particular zone of the virtual area. If the virtual area or zone currently is occupied by one or more communicants, the virtual area creator **18** establishes a presence for the PSTN terminal device user in the virtual area or zone and administers the connections of the sources and sinks of the communicants and the PSTN terminal device user through the VOiP gateway server **176**. If the virtual area is unoccupied (i.e., there is no one present), the virtual environment creator **18** instantiates a respective instance of the area service to administer the virtual area and establish a presence for the PSTN terminal device user in the virtual area or zone. In this case, the PSTN terminal device user may interact with data that is stored in association with the virtual area or zone, or may submit one or more commands (e.g., DTMF control characters) for the area service to request one or more communicants to join the PSTN terminal device user in the virtual area or zone.

[0119] FIG. **11A** shows an example **314** of the network communications environment **10** that

supports virtual area based telephony communications in which a user of a mobile device **316** (e.g., a mobile phone and a portable computing device, such as a tablet computer or a notebook computer) connects to a virtual area using a mobile client communications application.

[0120] The network communications environment **280** includes first and second area servers **172**, **174**, a VOiP gateway server **176**, a SIP proxy server **286**, a SIP registrar server **288**, and an optional map server **318**. Each of the first and second area servers **172**, **174** executes a respective instance of the area service that administers a respective virtual area in which communicants can interact with each other and with data stored in association with the virtual areas. In the illustrated example, the first area server **172** creates a first virtual area in which client network node users (represented by sprites **178**, **282**) are present. The second area server **174** creates a second virtual area in which a third client network node user (represented by sprite **284**) is present. The VOiP gateway server **176** communicates with the mobile device **316** and the SIP proxy server **286**. The SIP proxy server **286** communicates with the VOiP gateway server **176** and the SIP registrar server **288**. The map server **318** provides logon services to the client network nodes and the mobile device **316**.

[0121] In some examples, the mobile client communications application is a modified version of the virtual area based client applications **20**, **22** running on the client network nodes **12**, **14**. In some of these examples, the mobile client communications application includes a stripped down version of the SORK kernel described in U.S. application Ser. No. 12/630,973, filed on Dec. 4, 2009, that includes the message handling functions but omits the audio mixing functions.

[0122] FIG. **11B** is a flow diagram of an example virtual area based telephony method in which a user of a mobile device connects to a virtual area or a particular zone of a virtual area using a mobile client communications application. In this example, the mobile device **316** makes two separate connections: an audio connection **320** to the VOiP gateway server **176**; and a data connection **322** to the area server **172**.

[0123] In accordance with the method of FIG. **11B**, the mobile client communications application logs into the virtual environment creator **18** (FIG. **11B**, block **324**). In the illustrated example, the mobile client communications application logs in through the map server **318**, which selects an area server to administer the virtual area to which the mobile client communications application was requested to connect. The map server **318** sends the network address of the selected area server to the mobile client communications application.

[0124] The mobile client communications application negotiates a data connection **322** with the area service running on the selected area server (FIG. **11B**, block **326**). In some examples, the data is sent as SODA definitions in accordance with the STRAW stream transport protocol described in U.S. patent application Ser. No. 12/825,512, filed Jun. 29, 2010.

[0125] The mobile client communications application calls the SIP proxy server **286** with the user's user ID as the called telephone number (FIG. **11B**, block **328**). The SIP proxy server **286** requests the network address (e.g., a URI, such as a URL) of the VOiP gateway server associated with the user ID from the SIP registrar server **288** (FIG. **11B**, block **330**). The SIP registrar server **288** returns the network address of the VOiP gateway server to the SIP proxy server **286** (FIG. **10B**, block **332**).

[0126] The SIP proxy server **286** sends the SIP Invite message to the VOiP gateway server **176** (FIG. **11B**, block **334**). In response, the VOiP gateway server **176** sends a notification of the inbound telephone call to the area service hosting the virtual area associated with the user ID (FIG. **11B**, block **336**). The area services sends the session keys (also referred to as "credentials") for the virtual area to the VOiP gateway server **176** (FIG. **11B**, block **338**).

[0127] The VOiP gateway service completes the SIP Invite by sending a SIP Ack to the ITSP **192** (FIG. **11B**, block **342**). The VOiP gateway service and the ITSP **192** establish a RTP session on the RTP connection **320** (FIG. **11B**, block **344**). Audio data is transmitted between the ITSP **192**, the VOiP gateway service, and the gateway client instance over the RTP connection **320** (FIG. **11B**, block **346**). The audio data is transmitted between the gateway client instance and the client

network nodes (FIG. 11B, block 348). In this process, the gateway client instance mixes the realtime audio streams received from the other communicants in the virtual area to produce a single mixed audio stream that is transmitted to the mobile device 316.

[0128] In this example, the mobile client communications application renders a graphical user interface of the type shown in FIGS. 2-4 on a display of the mobile device 316. In this process, the mobile client application receives from the area service a specification of the virtual area, the positions of the objects in the virtual area, and realtime differential stream (RDS) data describing the states of the communicants' communication channels. Based on this data, the mobile client communications application renders on the display graphical representation of the communicants in spatial relation to a graphic representation of the virtual area, and visual cues indicating the states of the communicants' communications channels synchronized with the audio data received over the audio connection 320.

III. CONCLUSION

[0129] Other embodiments are within the scope of the claims.

Claims

1. A method comprising: connecting to a network communications environment supporting realtime communications between respective network nodes of a user and other communicants in virtual areas each of which is associated with its own respective set of communicant members; in connection with the user's network node, providing a graphical user interface comprising controls for establishing presence in respective ones of the virtual areas, managing realtime communications with other communicants in respective ones of the virtual areas, and presenting different views of communicants associated with the network communications environment; and based on user input in connection with the graphical user interface, establishing a presence for the user in a selected one of the virtual areas, administering realtime communications between the user and one or more communicants who are present in the selected virtual area, and displaying a visualization that shows graphical representations, locations of presence, and realtime activities of communicants across respective ones of the virtual areas.

2-43. (canceled)
